Knowing the unknown animal

Open-set recognition for camera-trap images at new sites

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animal images at a new site show a species the model was never trained on

11.3% across the four held-out test sites (2,690 of 23,800 animal images).

The problem

- Species classifiers are trained on a fixed list of species (a "closed set")
- At a new site, an animal not on the list still gets a name from the list
- In our 2025 pilot, 64% of unknown-species images got a known label with confidence above 0.9
- Volunteers then trust the label and skip the image
- Checking every image by hand: about 4.2 s per image for a trained volunteer

Data: TrailCam-Ostra

Images	412,000 animal images, 2019–2025
Known species	38 (mammals and ground birds)
Camera sites	61: 52 for training, 5 for validation, 4 for testing
Test sites	23,800 animal images; 2,690 show one of 14 species not among the 38
Empty frames	Removed beforehand with MegaDetector v5

Split by site, never by image: no camera appears in two splits.

Method (1): the energy score

$$E(x) = -T \cdot log\ \Sigma_k\ exp(f_k(x) / T)$$

- $f_k(x)$: the network's score (logit) for species k; T = 1
- Low energy: looks like a known species. High energy: unknown
- An image is flagged "unknown" when E(x) > τ
- Backbone: ViT-B/16 with DINOv2 weights, fine-tuned on the 52 training sites

Energy score after Liu et al. (2020).

Method (2): site-aware outlier exposure

$$L = L_{CE} + \lambda \cdot L_{OE}, \lambda = 0.5$$

- L_{CE}: the usual classification loss on the 38 known species
- L_{OE}: pushes energy up on outlier images: 31,000 iNaturalist photos of 300 species not among the 38
- Site-aware: each batch mixes outliers with known images from the same site, so the model cannot use the background to tell them apart
- Then temperature scaling, fitted on the 5 validation sites

Choosing the threshold τ

- τ is set on the 5 validation sites, never on the test sites
- Rule: keep 95% of known-species images (TPR = 95%)
- The same τ is then used unchanged at all 4 test sites

TPR: true positive rate, the share of known-species images accepted as known.

How we measure

- Top-1: accuracy on known species only
- AUROC: how well the score separates unknown from known (0.5 = chance, 1 = perfect)
- FPR@95: share of unknown animals still accepted as known, at TPR 95%. Lower is better
- ECE: expected calibration error, 15 bins:

$$ECE = \Sigma_b \, (n_b / N) \cdot | \, acc(b) - conf(b) \, |$$

Main results, 4 test sites

Method	Top-1 ↑	AUROC ↑	FPR@95 ↓	ECE ↓
Max softmax (baseline)	91.8 ± 0.3	81.2 ± 0.9	58.4 ± 2.1	6.1 ± 0.4
Energy score	91.8 ± 0.3	84.7 ± 0.7	49.0 ± 1.8	6.1 ± 0.4
Energy + outlier exposure	91.2 ± 0.4	90.3 ± 0.6	31.5 ± 1.5	4.8 ± 0.3
Ours	91.4 ± 0.3	**92.6 ± 0.5**	**24.7 ± 1.2**	**2.3 ± 0.2**

All numbers in %. Mean ± standard deviation over 5 training seeds.

Confidence you can take at its word

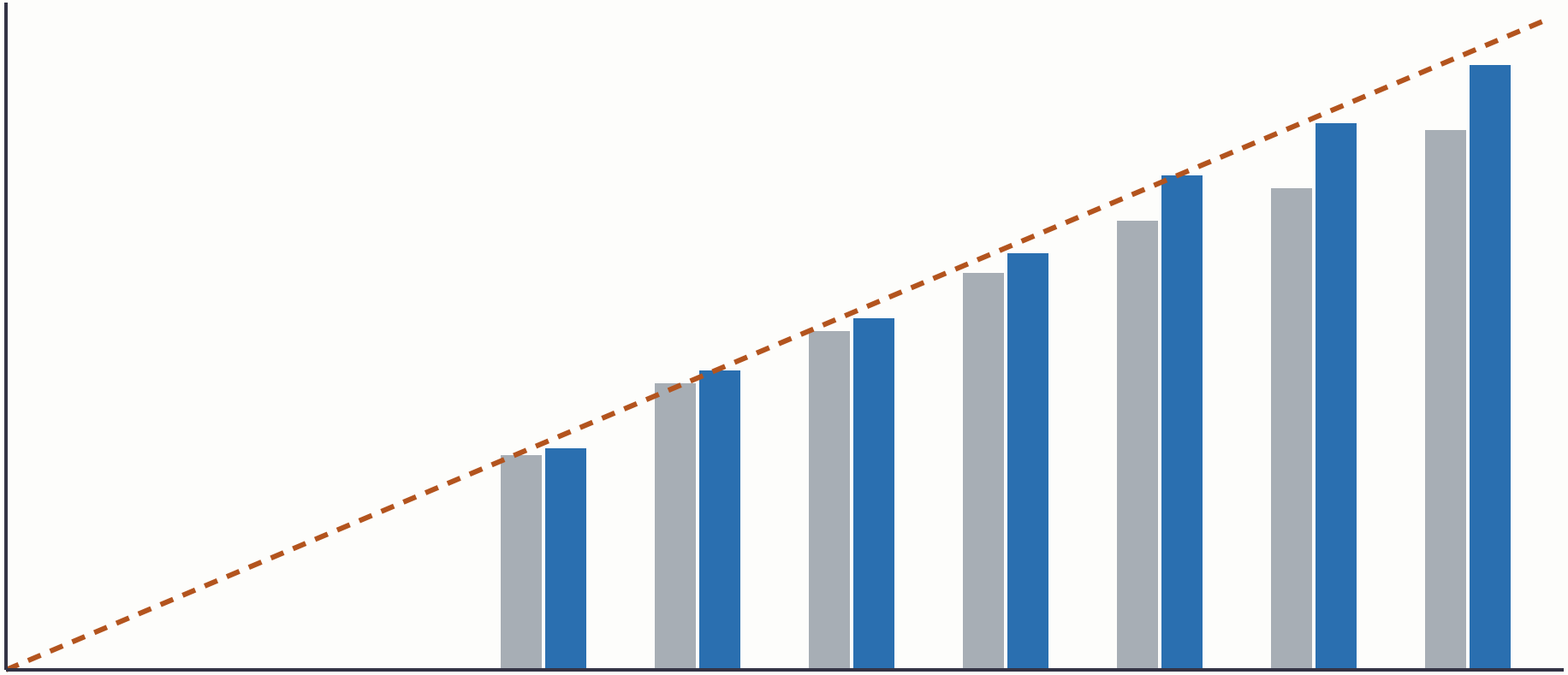


Figure 3. Reliability diagram, 4 test sites. Bars: accuracy in each confidence bin (15 bins; bins with fewer than 200 images not shown). Grey: max-softmax baseline. Blue: ours. Dashed: perfect calibration.

Site by site (ours)

Test site	Animal images	Unknown share	AUROC
A · lowland forest	7,900	6.2%	95.1
B · river edge	6,400	9.8%	93.4
C · upland heath	5,300	12.5%	92.0
D · conifer plantation, 81% night (IR)	4,200	21.6%	86.3

Mean of 5 seeds.

Where it still fails

- Night infrared images: AUROC 87.9, against 94.8 in daylight
- Small animals filling less than 5% of the frame
- 41% of the unknowns still accepted as known are rodents or shrews

A grey, blurred, small animal at night looks like every other grey, blurred, small animal at night.

Ablation (AUROC, mean of 5 seeds)

Variant	AUROC	ECE
Ours (λ = 0.5)	92.6	2.3
Without site-aware batches	90.9	2.5
Without temperature scaling	92.5	5.0
λ = 0.1	91.1	2.4
λ = 1.0	92.2	2.6

λ = 1.0 also lowers Top-1 to 90.1%.

What it means for volunteers

Per 10,000 animal images at a new site (11.3% unknown):

- About 1,300 are flagged and go to a person
- About 280 unknown animals still get a known name
- Review time: about 1.5 hours, instead of 11.7 hours for all 10,000

Compute

Model size	86 M parameters (ViT-B/16)
Training	14 h on 4 × A100 GPUs, per seed
Inference	38 ms per image on one A100, batch 64
Energy score	no extra cost: computed from the same logits

Limitations

- Four test sites, all in one temperate country
- "Unknown" means not among our 38 labels; most unknowns are close relatives of known species
- We removed outlier photos of the 14 test species, but not of their genera
- τ is fixed from validation sites; it may drift with season and camera model
- We flag images; we do not count individuals or estimate abundance

Conclusions

- Site-aware outlier exposure: AUROC 92.6 against 81.2 for the baseline, at almost the same Top-1
- Unknowns accepted as known: from 58% to 25%
- Confidence becomes honest: ECE from 6.1% to 2.3%
- Night and small animals remain the hard part

Code, data, funding

- Code: git.tellhaven.example/cv/unknown-animal (Apache 2.0)
- TrailCam-Ostra: images on request from the Ostra Wildlife Trust; site locations withheld to protect species
- Funded by the Tellhaven Biodiversity Fund, grant BF-24-112

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Thank you

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